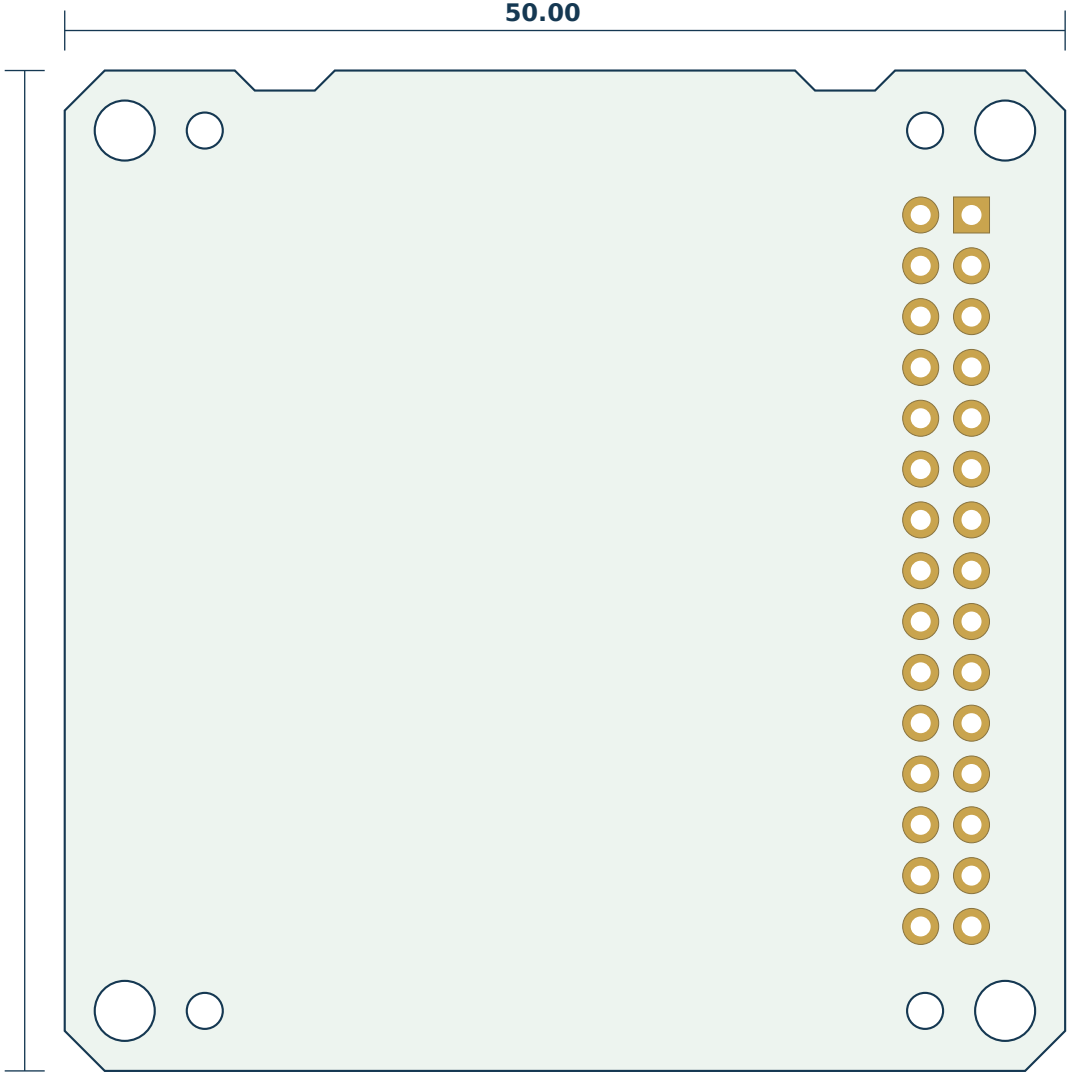
Fabrication drawing

Top view, toward CoreS3 SE. Dimensions in mm. Native Edge.Cuts and drill files control geometry.



Board thickness 1.60 nominal; 2.00 corner chamfers.

Four layers / JLC04161H-7628

L1 F.Cu: signals + ground, 0.035 mm copper

Prepreg: 0.2104 mm, nominal Dk 4.4

L2 In1.Cu: ground, 0.0152 mm copper

Core: 1.065 mm, nominal Dk 4.6

L3 In2.Cu: ground, 0.0152 mm copper

Prepreg: 0.2104 mm, nominal Dk 4.4

L4 B.Cu: components + RF, 0.035 mm copper

Fabrication requirements

ENIG; lead-free process; FR-4. Minimum design track / space 0.20 / 0.20. Signal traces 0.25; power 0.50. Vias 0.60 land / 0.30 finished drill. Soldermask expansion 0.05; minimum web 0.10. Tent vias both sides. J2 has an intentional merged mask opening; preserve its copper keepout.

RF ANT: 0.35 mm trace, 0.50 mm side clearance to B.Cu ground, referenced to solid L3. Fabricator must confirm 50 ohm +/-10% using the actual stack and soldermask. No copper or drill changes without design review.

Drills: 4 x 3.00 NPTH at (3,3), (47,3), (3,47), (47,47). 4 x 1.80 NPTH at (7,3), (43,3), (7,47), (43,47). J1: 30 x 1.00 PTH on 2.54 pitch with 1.80 lands; row X=42.78 / 45.32, first Y=7.22.

Quote tolerances: routed outline +/-0.10; finished holes +/-0.08; board thickness +/-0.10 mm. Confirm actual M032 seat and A089 pin fit before approving a production batch. Board-only STEP excludes component heights and the enclosure.

ENGINEERING FIRST ARTICLE - physical fit and native-radio operation not yet validated